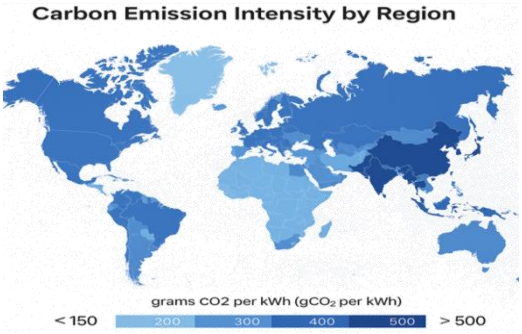
3.5 Power-to-Mint Engine

Turning Renewable Energy into Verifiable, Programmable Capital

The Power-to-Mint mechanism is AGV Protocol's signature engine for transforming real-world energy production into on-chain asset issuance. It forms a deterministic, auditable bridge between kilowatt-hour generation and GVT (Green Value Token) minting, secured by data provenance, smart contracts, and oracle validation.



Mechanism Overview

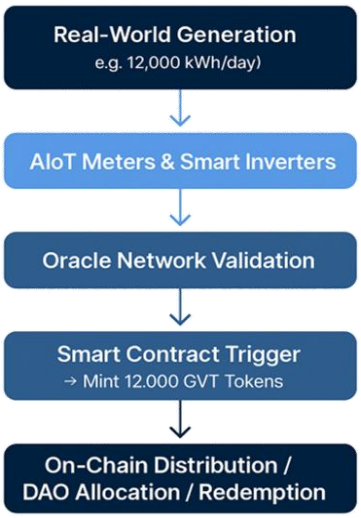
Component	Role	Description
Energy Asset (Solar, Hydro, Wind)	Source	On-grid and off-grid power generation units with real-time smart metering (e.g. photovoltaic farms, microgrids)
AIoT Metering	Data Layer	Devices log kWh generation with timestamped signatures; validated via multi-source redundancy
Oracle Aggregator	Verifier	Confirms device integrity and synchronizes yield data to minting logic
Minting Smart Contract	Token Logic	Mints GVT in 1:1 ratio to verified energy units (e.g., 1 kWh = 1 GVT) with daily/hourly settlement windows
Policy Metadata	Compliance	Energy source, carbon intensity, and jurisdictional policy attached as metadata

Power-to-Mint Flow

The Power-to-Mint mechanism represents AGV Protocol's deterministic link between real-world electricity production and token issuance. Each kilowatt-hour (kWh) of verified generation — captured via certified AIoT meters and validated by an oracle network — directly triggers the minting of 1 GVT token through a smart contract.

This process ensures:

- Proof of Generation: Only data with oracle-validated timestamps enters the minting logic.
- Auditability & Trust: All smart contracts are traceable and parameterized for ESG and policy compliance.
- Programmable Distribution: Minted GVT tokens are automatically routed into DAO treasuries, redemption pools, or staker wallets based on predefined rules.



By transforming clean energy into composable on-chain assets, AGV enables a capital-efficient infrastructure model where each unit of energy becomes a unit of programmable liquidity.

Advantages of Power-to-Mint

- Yield-Backed Minting

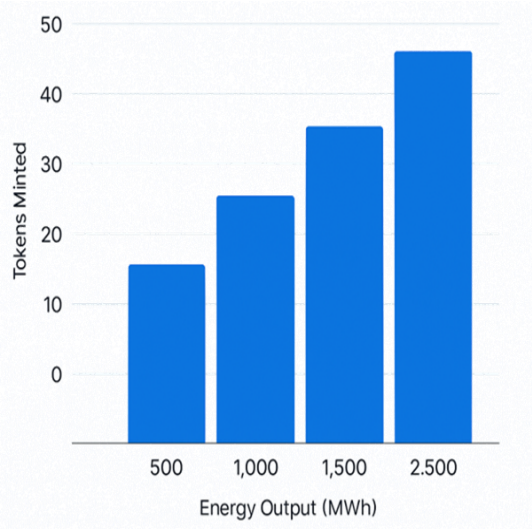
Tokens are backed by productive output—not speculation—ensuring sustainability and real economic value.

- Carbon-Aware Metadata

Each GVT can carry embedded environmental data (e.g., source carbon rating), enabling future integration with ESG and carbon credit systems.

- Policy Lending Integration

Allows programmable linkage between verified production and policy loans or subsidies (e.g., $ per kWh models).



- Tamper-Proof Distribution

Deterministic contracts eliminate manual intervention, enhancing trust with regulators and institutional partners.

- Programmable Supply Controls

Minting rate, cap, and burn rules are encoded per region, project, or DAO decision, ensuring controlled token emissions.